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Exercise: Day 5

1. Derive each of the equations boxed in red in the course lecture notes. Explain the physical significance of each equation.
2. In this problem you will consider the so-called Marcus inverted regime. Focusing only on the exponential term in the Marcus rate equation, $k_{\text{et}} \propto \exp(-\beta A^\ddagger)$, answer the following questions.
 - a. Show that the Marcus rate is maximized with $\Delta A = -\lambda$.
 - b. If the thermodynamic driving force for an electron transfer is ΔA , show that
 - i. If $|\Delta A| < \lambda$, then k_{et} increases as $|\Delta A|$ increases
 - ii. If $|\Delta A| > \lambda$, then k_{et} decreases as $|\Delta A|$ increases
 - c. Explain why case ii (the inverted regime) is counterintuitive from the perspective of transition state theory. What feature of the parabola crossing is responsible for its counterintuitive nature?